

Towards Transformer Winding Deformation Fault Diagnosis under Data Scarcity: Integrating Frequency Response Analysis with Metric Learning

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Abstract—Accurate diagnosis of transformer winding deformation faults is critical for ensuring the reliability and stability of power systems. However, the scarcity of real-world faulty frequency response analysis (FRA) data severely constrains the development of intelligent data-driven diagnostic models. To overcome this limitation, this study proposes a metric learning framework that integrates FRA with triplet loss and virtual adversarial training (VAT). The method first reformulates the original dataset of N FRA samples into N^2 training pairs through the random pairing mechanism, thereby substantially expanding the effective training set. Triplet loss is then applied to further enrich sample diversity ($\approx N^3$) and enhance model feature extraction. To address the inherent label imbalance arising from random pairing, VAT is incorporated. This regularization technique generates synthetic intra-class pairs and enforces local smoothness in the embedding space. Experimental validation on a specially designed 10 kV transformer demonstrates that the proposed approach achieves superior diagnostic accuracy. When applied to two practical transformers with confirmed winding deformations, the trained model accurately identifies fault types. Importantly, the method offers a potential solution for detecting previously unknown winding faults.

Index Terms—Data scarcity, fault diagnosis, frequency response analysis, metric learning, transformer, winding deformation fault.

This work was supported in part by the China Southern Power Grid under Grant No. ZBKJXM20232293, and in part by the Fundamental Research Funds for the Central Universities under Grant SWU-KT22027 (*Corresponding author: Yuxuan Ding and Yaping Du*).

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I. INTRODUCTION

POWER transformers are vital to power system operation, providing the interface between different voltage levels and directly influencing reliability and stability of the power system [1]–[3]. However, with the rapid expansion of power system capacity, the instantaneous short-circuit (SC) current has markedly increased, making transformers more vulnerable to faults. Considering their substantial cost, timely fault detection is imperative. Statistics from the International Council on Large Electric Systems (CIGRE) indicate that winding deformation accounts for nearly 30% of transformer failures [1], [4], [5]. While minor deformation may not immediately disrupt normal operation, its progression can ultimately cause catastrophic failure if neglected. Therefore, when transformers are disconnected from the grid, accurate diagnostic methods for winding deformation during periodic inspections are essential.

Several diagnostic methods have been developed to detect transformer faults, each based on distinct measurement principles. For instance, optical fiber sensing methods employ Fabry–Pérot (F-P) or Fiber Bragg Grating (FBG) sensors to capture ultrasonic signals and mechanical vibrations for diagnosing partial discharges in transformers [6]–[8]; dissolved gas analysis (DGA) assesses transformer insulation status by interpreting the concentrations and compositions of combustible gases dissolved in oil [9]; vibration-based approaches extract characteristic signals associated with winding faults [10], [11]; short-circuit impedance method evaluates variations in resistance and inductance components to diagnose winding deformation [2], [12]; voltage–current diagram techniques detect winding deformation by analyzing variations in voltage and current signals [13]–[15]; and frequency response analysis (FRA) method characterizes winding faults by mapping them to variations in the amplitude–frequency response [16]–[18]. Each of these methods offers distinct strengths and limitations, thereby motivating the development of diverse diagnostic approaches. Among them, FRA is the most widely adopted and well-established technique due to its high sensitivity, accuracy, and ease of application [19]–[21]. Beyond winding faults, FRA has also been employed to diagnose bushing defects and insulation oil degradation [17]. However, in rou-

tine transformer inspections, the interpretation of FRA data still depends heavily on subjective expert judgment, thereby limiting diagnostic accuracy to the scope of individual domain expertise.

Currently, three primary approaches are employed to objectively interpret FRA data: (1) constructing transformer equivalent circuit models to elucidate winding fault mechanisms; (2) developing mathematical–statistical indicators with expert-defined thresholds for fault diagnosis; and (3) applying data-driven methods to classify winding fault types.

In the first category, Refs. [19], [22] construct the equivalent circuits of transformer windings and derive their parameters using the finite element method (FEM), thereby providing a theoretical foundation for interpreting axial displacement (AD) and radial deformation (RD) faults. Ref. [16] integrates a gray-box winding model with FEM to simulate inter-disk short circuit (IDSC) and disk space variation (DSV) faults, enhancing the understanding of fault mechanisms. Beyond FEM-based parameter estimation, Refs. [23]–[26] integrate various optimization algorithms with gray-box models to perform simulations in either the time or frequency domain. Overall, this modeling approach requires substantial prior knowledge of transformer structure and considerable effort to construct the equivalent circuit, thereby limiting its practicality for FRA data interpretation.

In the second category, Refs. [27], [28] introduce many mathematical–statistical indicators with expert-defined thresholds to evaluate the type and severity of winding faults using standard amplitude–frequency response. Representative indicators include the root mean square error (RMSE), correlation coefficient (CC), and absolute sum logarithmic error (ASLE). Although the values of these indicators vary across different transformers, their changing trends under identical fault conditions remain consistent [28]. Among them, CC is the most widely adopted and is endorsed by the Chinese power industry standard DL/T 911-2016 [29], which provides practical guidance for its application. Besides, Refs. [30], [31] extend the analysis by constructing polar plots that integrate both amplitude- and phase-frequency responses, followed by application of corresponding indicators to interpret FRA data. Overall, these indicator-based methods rely on predefined and subjective thresholds, rendering fault diagnosis dependent on expert judgment.

In the third category, Refs. [32], [33] employ various classifiers, such as support vector machines (SVMs), isolation forest, and k-means, based on predefined indicators to enhance fault diagnosis. In contrast, other Refs. [34], [35] directly utilize FRA sequence or image data as neural network (NN) inputs for end-to-end fault diagnosis. Traditional machine learning (ML) approaches typically operate on small-scale datasets (≈ 100), whereas deep learning (DL) methods exploit larger datasets (≈ 1000). Nevertheless, real-world labeled faulty FRA data remain extremely scarce due to commercial confidentiality of transformer manufacturers and utilities, resulting in the absence of publicly accessible datasets [1]. Consequently, most datasets in existing studies are derived from laboratory winding deformation experiments, which are time-consuming, costly, and risky [1], [32], [33]. Notably, disassembling and

reassembling a transformer to replace faulty windings requires approximately one day, whereas collecting a dataset of merely 100 samples for a single transformer may take up to three months. Overall, this persistent data scarcity constitutes a fundamental barrier to advancing reliable, data-driven intelligent fault diagnosis for power transformers.

For data-driven methods, the scarcity of transformer fault data remains a critical challenge. To address this limitation, three primary strategies have been explored. The first employs physics-based models to generate synthetic fault data [16]. However, these models often lack sufficient maturity, producing simulated data that diverges from real measurements in terms of physical fidelity and consistency. The second strategy leverages transfer learning [36], [37], where models pretrained on large-scale datasets are fine-tuned on limited data. While effective in reducing data requirements, this approach presupposes access to substantial pretraining data, which is not always feasible. The third strategy utilizes generative artificial intelligence (AI) techniques [1], such as generative adversarial networks (GANs) or diffusion models, to augment datasets by synthesizing additional fault samples. Nevertheless, these methods exhibit inherent black-box characteristics, introduce uncertainty in data quality, and typically require large real datasets to achieve stable and reliable training.

To address these challenges, and inspired by Ref. [38], this study proposes a metric learning framework that integrates FRA to detect winding deformation faults under data scarcity. The main contributions are summarized as follows:

- 1) We propose a metric learning framework integrating with FRA by reformulating the original dataset of N samples into N^2 training pairs. This random pairing mechanism substantially expands the effective training set, thereby facilitating NN training under data scarcity.
- 2) We employ triplet loss to exploit diverse sample combinations. This strategy further augments dataset variability ($\approx N^3$) while simultaneously minimizing intra-class variations and maximizing inter-class separation in the learned embedding space, thereby enhancing feature discriminability.
- 3) We incorporate virtual adversarial training (VAT) to mitigate label imbalance where inter-class pairs vastly outnumber intra-class pairs. This regularization technique effectively augments intra-class pairs and promotes local smoothness in the embedding space, improving model robustness.

The remainder of this study is organized as follows: Section II introduces the principles of FRA, metric learning, triplet loss, and VAT. Section III describes the experimental dataset. Section IV presents the experimental results. Finally, Section V concludes this study.

II. METHODOLOGY

This section first introduces the principles of FRA-based transformer winding deformation diagnosis, followed by the fundamentals of metric learning and its integration with triplet loss. Finally, the theory of VAT is introduced. The overall framework of the proposed method is depicted in Fig. 1.

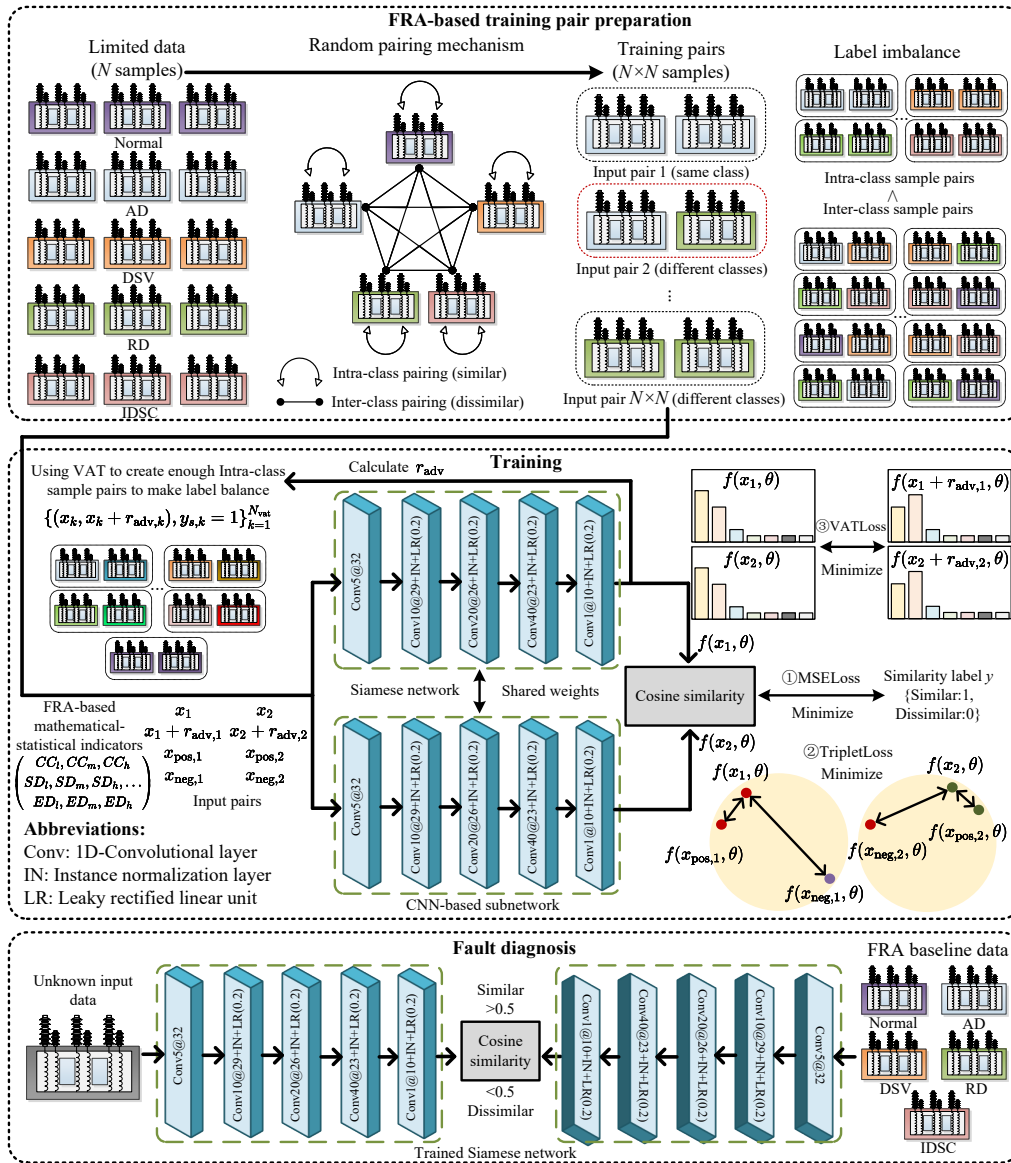


Fig. 1. Framework of the proposed method. During FRA-based training pair preparation, N FRA samples are randomly paired to N^2 training pairs, resulting in more inter-class than intra-class pairs. To address this label imbalance, VAT is employed to generate sufficient intra-class pairs and enhance model robustness. In addition, triplet loss further enriches dataset diversity ($\approx N^3$) and strengthens feature extraction. In the fault diagnosis stage, the trained Siamese network assesses the similarity between input and baseline data to detect winding faults.

A. Fundamentals of FRA and Associated Indicators

Under high-frequency excitation (typically above 1 kHz), the transformer core exhibits negligible magnetizing effects, enabling the winding to be modeled as a passive linear two-port network characterized by distributed parameters, including resistance, inductance, and capacitance [16], [20], [29]. In standard FRA tests, a sinusoidal input voltage $\vec{R}_{in}(\omega)$ is applied at one terminal across a frequency sweep (commonly 1 kHz to 1 MHz [29]), while the output response $\vec{R}_{out}(\omega)$ is measured at the other terminal. The resulting transfer function $T(\omega)$, commonly referred to as the FRA fingerprint, FRA signature, or FRA data, is obtained as defined in Equation (1). This fingerprint uniquely characterizes the winding condition, whether healthy or deformed, as the winding's geometric configuration inherently determines the distributed parameters.

Mechanical deformations alter these parameters, inducing shifts in resonance and anti-resonance frequencies and thereby modifying the overall FRA data. By comparing measured FRA data against a baseline (healthy) fingerprint, maintenance personnel can analyze these variations to assess the winding condition [20].

$$T(\omega) = 20 \log_{10} \left| \frac{\vec{R}_{out}(\omega)}{\vec{R}_{in}(\omega)} \right| \text{ dB} \quad (1)$$

where $\vec{R}_{in}(\omega)$ and $\vec{R}_{out}(\omega)$ are excitation and response signals, and $T(\omega)$ is the frequency response fingerprint (amplitude versus frequency).

Building upon previous studies [20], [28], this study employs mathematical-statistical indicators as model inputs rather than directly utilizing FRA sequences or image data.

This choice is motivated by two key considerations [1]. First, these indicators effectively capture differences between the measured FRA data and the reference fingerprint. Although FRA data vary across different transformers, the changing trends of these indicators remain consistent. This consistency enhances model generalization and alleviates the transformer-specific biases inherent in unprocessed raw-data approaches. Second, the dimensionality of the indicators is substantially lower than that of FRA sequences or image data, enabling the deployment of lightweight NNs with reduced computational demands, faster convergence, and lower susceptibility to overfitting in data-scarce regimes [1], [35]. This study adopts several commonly used mathematical-statistical indicators, with their definitions provided in Eqs. (2)–(12). For further details, readers could refer to Refs. [1], [27], [28].

$$CC = \frac{\sum_{\omega=1}^N T_m(\omega)T_r(\omega)}{\sqrt{\sum_{\omega=1}^N |T_m(\omega)|^2 \sum_{\omega=1}^N |T_r(\omega)|^2}} \quad (2)$$

$$SD = \sqrt{\frac{\sum_{\omega=1}^N (T_m(\omega) - T_r(\omega))^2}{N - 1}} \quad (3)$$

$$ASLE = \frac{1}{N} \sum_{\omega=1}^N |20 \log_{10} |T_r(\omega)| - 20 \log_{10} |T_m(\omega)|| \quad (4)$$

$$SSRE = \frac{1}{N} \sum_{\omega=1}^N \left(\frac{T_m(\omega)}{T_r(\omega)} - 1 \right)^2 \quad (5)$$

$$Rou = \frac{\sum_{\omega=1}^N \tilde{T}_r(\omega) \tilde{T}_m(\omega)}{\sqrt{\sum_{\omega=1}^N |\tilde{T}_r(\omega)|^2} \sqrt{\sum_{\omega=1}^N |\tilde{T}_m(\omega)|^2}} \quad (6)$$

$$MM = \frac{\sum_{\omega=1}^N \min(|T_r(\omega)|, |T_m(\omega)|)}{\sum_{\omega=1}^N \max(|T_r(\omega)|, |T_m(\omega)|)} \quad (7)$$

$$E[\Delta] = \frac{1}{N} \sum_{\omega=1}^N \Delta(\omega), \quad \Delta(\omega) = \frac{T_m(\omega) - T_r(\omega)}{\frac{1}{N} \sum_{\omega=1}^N T_r(\omega)} \quad (8)$$

$$\Delta = \frac{1}{N} \sum_{\omega=1}^N \sqrt{\left(\frac{T_r(\omega) - a/2}{a} \right)^2 + \left(\frac{T_m(\omega) - a/2}{a} \right)^2} \quad (9)$$

$$a = T_r(\omega) + T_m(\omega) \quad (10)$$

$$DABS = \frac{1}{N} \sum_{\omega=1}^N |T_r(\omega) - T_m(\omega)| \quad (11)$$

$$ED = \sqrt{\sum_{\omega=1}^N (T_m(\omega) - T_r(\omega))^2} \quad (12)$$

where ω is frequency index and N is number of sampling points. $T_r(\omega)$ and $\tilde{T}_r$ represent the reference FRA fingerprint and its normalized form for a healthy winding, respectively, while $T_m(\omega)$ and $\tilde{T}_m$ correspond to those of the measured winding. Following prior studies [1], [29], this study divides the FRA data into three sub-frequency bands: 1–100 kHz (low-frequency), 101–600 kHz (mid-frequency), and 601–1000 kHz (high-frequency). The mentioned mathematical-statistical indicators are computed within each sub-frequency band and

subsequently normalized, yielding a 30-dimensional input data after frequency division, as shown below.

$$x = [CC_l, CC_m, CC_h, \dots, ED_l, ED_m, ED_h] \quad (13)$$

where the subscripts l , m , and h represent the low-, middle-, and high-frequency bands, respectively.

B. Combination of Metric Learning with Triplet Loss

Typically, the FRA dataset for a single transformer contains approximately 100 samples [1], [32], [33], which is insufficient for training the NN-based models. In the context of building heating, ventilation, and air conditioning (HVAC) fault diagnosis, Ref. [38] addresses this issue by employing metric learning with random sample pairing to expand the effective training dataset. Building upon this idea, this study further integrates triplet loss into the metric learning framework.

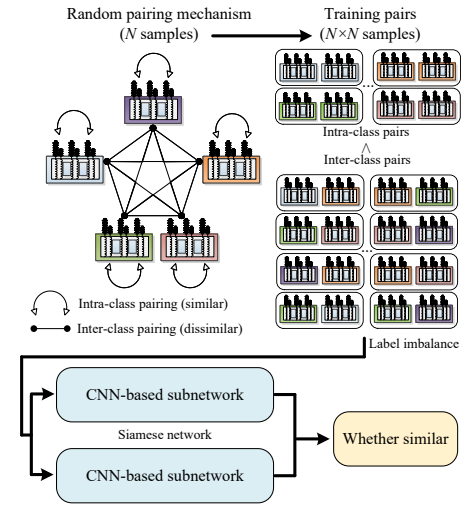


Fig. 2. Schematic representation of metric learning with the random pairing mechanism. Input pairs are processed by the Siamese network, and similarity is evaluated using cosine similarity. The Siamese network consists of two shared-weight CNN-based subnetworks in this study.

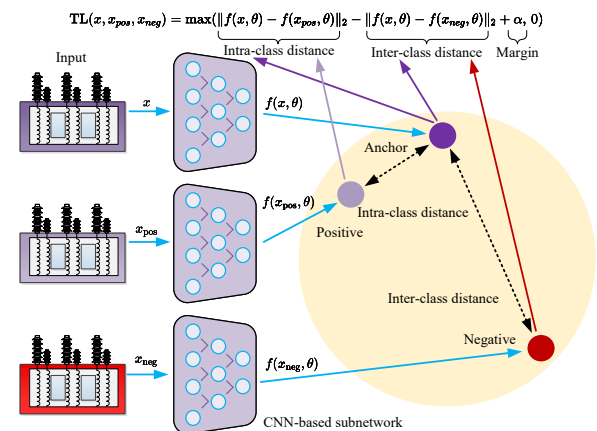


Fig. 3. Schematic illustration of the triplet loss applied to a sample.

Given a dataset $\mathcal{D}_{\text{origin}} = \{(x^i, y^i)\}_{i=1}^N$ consisting of N FRA-derived indicators x_i and their corresponding fault labels y_i , the objective of metric learning is to train a Siamese

network (consisting of two shared-weight NN-based models $f(\cdot)$) that projects input samples into a feature space where intra-class similarity is maximized and inter-class similarity is minimized, thereby enabling similarity assessment between samples.

$$\|f(x^i) - f(x^j)\|_2 \ll \|f(x^i) - f(x^k)\|_2 \quad \text{when } y^i = y^j \neq y^k \quad (14)$$

where x^i, x^j are samples from the same class, x^k is a sample from a different class, and $\|\cdot\|_2$ is the L2 norm. In other words, metric learning reformulates the task as pairwise comparisons, as illustrated in Fig. 2. Each sample is randomly paired with all others to generate N^2 training pairs, where each pair (x_1^i, x_2^i) is labeled as similar ($y_s = 1$, if $y^i = y^j$) or dissimilar ($y_s = 0$, $y^i \neq y^j$). Through this random pairing mechanism, the original dataset $\mathcal{D}_{\text{origin}}$ containing N FRA samples is expanded into an augmented dataset $\mathcal{D} = \{(x_1^i, x_2^i, y_s^i)\}_{i=1}^{N^2}$ containing N^2 training pairs, thereby substantially increasing the effective training set size. The loss function of metric learning can be defined as follows:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N^2} \sum_{i=1}^{N^2} (\text{cossim}[f(x_1^i, \theta), f(x_2^i, \theta)] - y_s^i)^2 \quad (15)$$

where $\text{cossim}[\cdot, \cdot]$ is cosine similarity. Relying solely on Eq. (15) as the loss function may cause the NN to memorize whether training pairs are similar or dissimilar, without learning meaningful feature representations. In other words, it fails to enforce the principle that features of the same class should be closer, while those of different classes should be farther apart. To address this limitation, this study introduces triplet loss, which minimizes intra-class variations while maximizing inter-class separation, as shown in Fig. 3. Triplets are formed by selecting an anchor sample x , a positive sample x_{pos} (same class as x), and a negative sample x_{neg} (different class to x). The triplet loss is defined as follows:

$$\begin{aligned} \text{TL}(x, x_{\text{pos}}, x_{\text{neg}}) &= \max\left(\|f(x, \theta) - f(x_{\text{pos}}, \theta)\|_2 \right. \\ &\quad \left. - \|f(x, \theta) - f(x_{\text{neg}}, \theta)\|_2 + \alpha, 0\right) \quad (16) \\ \mathcal{L}_{\text{TL}} &= \frac{1}{N^2} \sum_{i=1}^{N^2} [\text{TL}(x_1^i, x_{\text{pos},1}^i, x_{\text{neg},1}^i) + \text{TL}(x_2^i, x_{\text{pos},2}^i, x_{\text{neg},2}^i)] \quad (17) \end{aligned}$$

where α is a positive margin parameter (1 in this study). Moreover, by dynamically exploring diverse triplet combinations (potentially up to N^3), the approach introduces additional variability, mitigates overfitting, and captures subtle fault-invariant patterns in FRA-derived indicators.

C. Fundamentals of VAT for Addressing Label Imbalance and Enhancing Robustness

As illustrated in Fig. 2, metric learning inherently leads to label imbalance, as the number of samples within any single class is significantly smaller than the total samples in all other classes. Consequently, random pairing produces far more inter-class pairs than intra-class pairs. Unlike Ref. [38], which leaves this issue unaddressed, the proposed framework

mitigates the label imbalance by incorporating VAT (see Fig. 1). This approach not only augments effective intra-class pairs but also enhances model robustness.

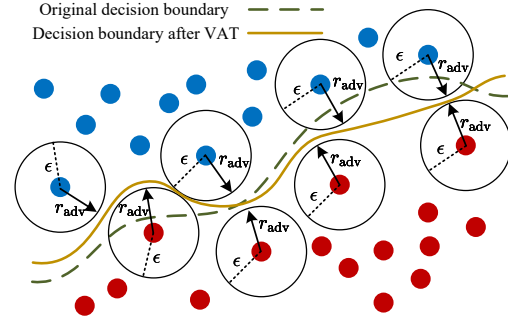


Fig. 4. Decision boundary before and after using VAT.

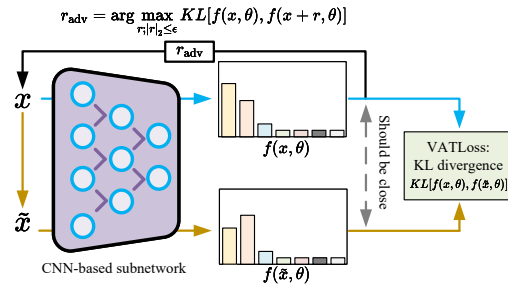


Fig. 5. Calculation process of VAT loss.

VAT often serves as an effective regularization technique to enhance model robustness by promoting local smoothness in the decision boundary within the embedding space [39], as shown in Fig. 4. Unlike conventional adversarial training, which relies on ground-truth labels to induce misclassification, VAT operates in a label-free manner. Specifically, it minimizes the divergence between the model's predictive distributions for the original input and its adversarially perturbed counterpart (see Fig. 5), thereby promoting consistency. Formally, for an input sample x , the VAT loss is defined as:

$$\text{VAT}(x) = KL[f(x, \theta), f(x + r_{\text{adv}}, \theta)] \quad (18)$$

$$r_{\text{adv}} = \arg \max_{r; |r|_2 \leq \epsilon} KL[f(x, \theta), f(x + r, \theta)] \quad (19)$$

$$\mathcal{L}_{\text{VAT}} = \frac{1}{N^2} \sum_{i=1}^{N^2} [\text{VAT}(x_1^i) + \text{VAT}(x_2^i)] \quad (20)$$

where $KL[\cdot, \cdot]$ denotes the Kullback-Leibler divergence, ϵ is a small perturbation bound (0.01 in this study), and r_{adv} represents the adversarial perturbation. By enforcing prediction consistency under such perturbations, VAT achieves two complementary effects: (1) it increases the number of effective intra-class pairs by generating perturbed samples that remain within the same class, thereby alleviating label imbalance; (2) it enhances robustness by promoting local smoothness in the embedding space, reducing sensitivity to noise, and improving generalization.

Algorithm 1 Proposed Metric Learning Framework Integrating with FRA

Require: Original dataset $\mathcal{D}_{\text{origin}}$ with N samples; Siamese network $\mathcal{M}$ composed of two shared-weight NN-based models f ; optimizer O ; number of epochs N_{epoch} ; hyperparameters $\alpha_{\text{TL}}, \alpha_{\text{VAT}}$

Ensure: Optimized Siamese network $\mathcal{M}$ with parameters θ

- 1: **Step 1: Generate Training Pairs**
- 2: Randomly pair each sample in $\mathcal{D}_{\text{origin}}$ with all others to construct $\mathcal{D}$ containing N^2 pairs.
- 3: Label each pair (x_1^i, x_2^i) as similar ($y_s^i = 1$ if $y_1^i = y_2^i$) or dissimilar ($y_s^i = 0$ if $y_1^i \neq y_2^i$).
- 4: **for** $epoch = 1$ to N_{epoch} **do**
- 5: **Step 2: Address Label Imbalance via VAT**
- 6: **for** each batch (x_1^i, x_2^i, y_s^i) **do**
- 7: Count intra-class pairs N_{pos} and inter-class pairs N_{neg}
- 8: **if** $N_{\text{neg}} > N_{\text{pos}}$ **then**
- 9: $N_{\text{vat}} = N_{\text{neg}} - N_{\text{pos}}$
- 10: **for** $k = 1$ to N_{vat} **do**
- 11: Randomly select a sample x_p from $\mathcal{D}_{\text{origin}}$
- 12: Generate adversarial perturbation r_{adv}
- 13: Construct augmented sample $\tilde{x}_k = x_k + r_{\text{adv},k}$
- 14: Add $(x_k, \tilde{x}_k, y_{s,k} = 1)$ into the batch
- 15: **end for**
- 16: **end if**
- 17: **Step 3: Compute Loss and Update Parameters**
- 18: Obtain embeddings $f(x_1^i, \theta), f(x_2^i, \theta)$ from $\mathcal{M}$
- 19: Compute MSE loss $\mathcal{L}_{\text{MSE}}$ (Eq. 15)
- 20: Compute Triplet loss $\mathcal{L}_{\text{TL}}$ (Eq. 17)
- 21: Compute VAT loss $\mathcal{L}_{\text{VAT}}$ (Eq. 20)
- 22: Aggregate total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \alpha_{\text{TL}}\mathcal{L}_{\text{TL}} + \alpha_{\text{VAT}}\mathcal{L}_{\text{VAT}}$
- 23: Backpropagate gradients w.r.t. $\mathcal{L}_{\text{total}}$ and update parameters θ via O
- 24: **end for**
- 25: **end for**

In summary, the total loss of the proposed framework is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \alpha_{\text{TL}}\mathcal{L}_{\text{TL}} + \alpha_{\text{VAT}}\mathcal{L}_{\text{VAT}} \quad (21)$$

where α_{TL} and α_{VAT} are hyperparameters that balance the contributions of different loss functions. The overall framework of the proposed method is illustrated in Fig. 1, with the detailed procedure presented in Algorithm 1. After training, the measured FRA data are compared against a library of pre-stored reference signatures representing various winding deformation faults through the Siamese network, enabling fault diagnosis by evaluating similarity. This procedure is fully consistent with the conventional FRA graphical comparison method [20].

III. FRA-BASED TRANSFORMER WINDING DEFORMATION FAULT EXPERIMENT

This section introduces the FRA dataset used for training and provides a brief description of the artificially simulated winding deformation faults.

TABLE I

DIMENSIONAL PARAMETERS OF THE EXPERIMENTAL TRANSFORMER

Parameter	Value
Iron core diameter (mm)	300
Iron core yoke length (mm)	1390
Iron core yoke height (mm)	1190
Tank dimensions (mm)	1705 × 740 × 1415
Number of disks	30
Number of turns per disk	10
HV & LV winding inner radius (mm)	421 & 316
HV & LV winding outer radius (mm)	500 & 349
HV & LV winding height (mm)	520 & 87.5

An experimental platform is established using a specially designed 10 kV transformer, whose nameplate parameters are provided in Table I and further detailed in Refs. [16], [21]. The FRA tests follow the end-to-end open-circuit connection as specified in DL/T 911-2016 [29], with the schematic shown in Fig. 6a, and the physical laboratory setup in Fig. 6b. All measurements are performed using a commercial FRA analyzer (Model: TDT6U, Beijing Shengtai Real-Time Electric Technology Co.), with specifications including: maximum output 25 Vpp (adjustable), output impedance 50 Ω , input impedance 1 M Ω /50 Ω (optional), frequency sweep range 20 Hz–2 MHz, frequency accuracy 0.005%, dynamic range –120 dB to 20 dB, gain accuracy ± 0.5 dB, and test time less than 1 min (1–1000 kHz/1000 points).

As shown in Fig. 7, four representative winding deformation faults, including DSV, IDSC, RD, and AD, are artificially simulated on the built platform [1], [40]. The DSV fault is simulated by connecting conductors of several disks with parallel capacitors, where the ratio of the added capacitance to the equivalent disk capacitance (2nf) quantifies fault severity. IDSC faults are introduced by shorting conductors between different disks; for instance, the notation “#1–#2” denotes that the first and second conductors of the middle sequential windings, viewed from top to bottom, are short-circuited. RD faults are simulated by manufacturing windings with varying degrees and directions of radial deformation, as illustrated in Fig. 7, where r is the winding radius, d is the radial displacement, and θ is the angular position of the deformed section. The severity of RD faults is expressed by the ratio d/r . AD faults are simulated by vertically shifting the high-voltage (HV) winding relative to the low-voltage (LV) winding, resulting in mechanical misalignment and altering the distribution of capacitance and inductance. The severity of AD faults is quantified by the ratio of axial displacement length Δh to total winding height h . Fig. 8 shows typical FRA data under different winding deformation faults. The dataset comprises 66 samples in total, including 5 AD faults, 21 DSV faults, 15 IDSC faults, 18 RD faults, and 7 normal cases. Further details are provided in Refs. [1], [21], [28]. Based on the random pairing mechanism, 4,356 pairs are generated, with 200 allocated to the test set and 4,156 to the training set.

IV. EXPERIMENTAL RESULTS

This section first conducts ablation studies to evaluate the contribution of individual framework components. Sub-

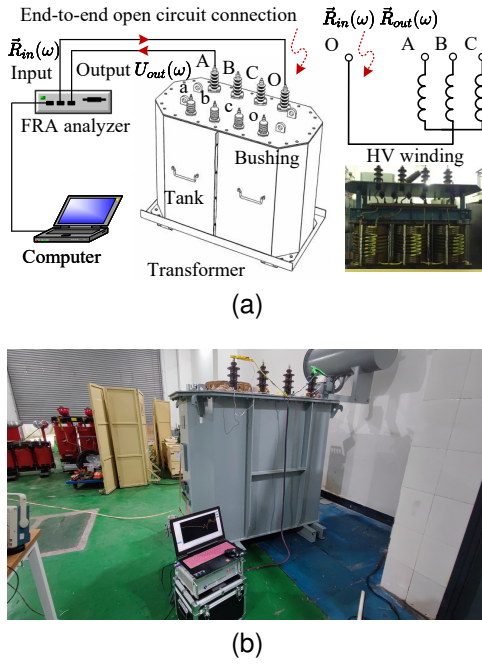


Fig. 6. Measurement experimental diagram. (a) Measurement wiring diagram. (b) Actual wiring diagram.

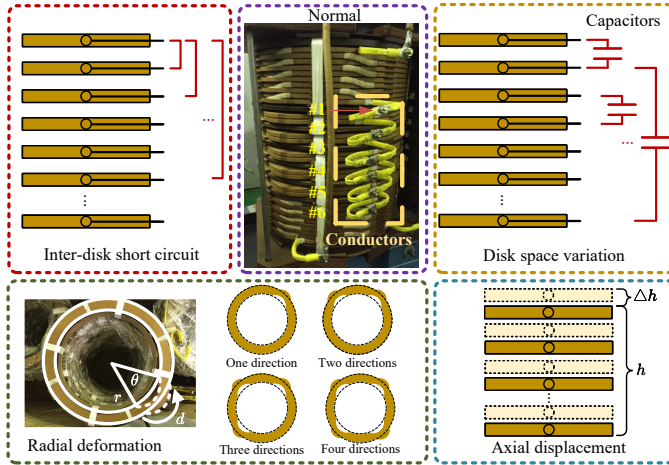


Fig. 7. Schematic representation of winding deformation faults.

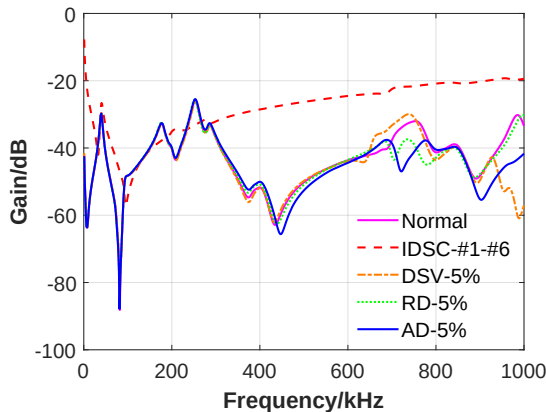


Fig. 8. Typical FRA data under different winding deformation faults.

TABLE II
COMPUTER HARDWARE AND SOFTWARE CONFIGURATION

Device or software	Nameplate value or configuration
CPU	Intel Ultra 7 265KF
GPU	RTX 5060 Ti
RAM	32G
Software	Python
Package	PyTorch, Numpy, Pandas, Sklearn

recorded data from two faulty transformers. Finally, a comparative analysis with conventional methods is presented, highlighting the distinct advantages of the proposed approach.

All performance metrics reported in this section, including accuracy, area under the curve (AUC), F1-score, precision, and recall, are evaluated on the test set. The experiments are conducted within the computational environment specified in Table II, and the architecture of the proposed NN-based model is illustrated in Fig. 1. The complete source code, together with hyperparameter settings, is publicly available in the associated GitHub repository¹.

A. Ablation Studies of Model Components

This study systematically compares the performance of Siamese networks constructed with different subnetworks, including convolutional neural networks (CNN), fully connected networks (FCN), recurrent neural networks (RNN), gated recurrent units (GRU), long short-term memory (LSTM), and transformer encoders. As shown in Table III, the CNN-based subnetwork achieves perfect performance across all metrics with minimal training time, confirming its effectiveness and efficiency. In contrast, RNN- and LSTM-based subnetworks exhibit lower accuracy and longer training times, while GRU- and transformer-based subnetworks provide competitive results but at a higher computational cost. Moreover, instance normalization (IN) proves to be an effective technique for enhancing performance [1], [35].

Table IV summarizes the performance of various strategies for mitigating label imbalance. Without any balancing (i.e., None), the model achieves an accuracy of approximately 0.92, reflecting the bias toward inter-class pairs due to their dominance in the dataset. The VAT strategy effectively mitigates label imbalance by generating additional intra-class pairs, thereby achieving superior performance across all evaluation metrics and demonstrating its critical contribution to the framework. In contrast, oversampling intra-class pairs and undersampling inter-class pairs improve accuracy to around 0.98. However, neither strategy achieves the performance attained by VAT, indicating that conventional label balancing strategies are less effective than adversarial augmentation in enhancing model generalization.

Table V illustrates the accuracy variations under different values of α_{VAT} and α_{TL} . The results indicate that setting both parameters within the range of 0.2–0.4 achieves optimal performance, suggesting that the combination of VAT loss and triplet loss synergistically enhances model generalization. In

sequently, the model's generalization is evaluated using field-

¹https://github.com/cyl034429432/FRA_Metric_Learning

TABLE III
SIAMESE NETWORK PERFORMANCE WITH DIFFERENT SUBNETWORKS

Subnetwork	Accuracy	AUC	F1-score	Precision	Recall	Training time
CNN-based subnetwork	1.00	1.00	1.00	1.00	1.00	8 min
CNN-based subnetwork without instance norm	0.98	0.99	0.97	0.97	0.99	8 min
FCN-based subnetwork	0.89	0.97	0.90	0.82	1.00	20 min
RNN-based subnetwork	0.82	0.92	0.83	0.77	0.91	51 min
GRU-based subnetwork	0.94	0.96	0.94	0.91	0.97	55 min
LSTM-based subnetwork	0.77	0.90	0.80	0.71	0.91	42 min
Transformer-based subnetwork	0.93	0.98	0.93	0.94	0.93	264 min

contrast, extreme settings (e.g., $\alpha_{\text{VAT}} = 0$ or 0.8) lead to slight accuracy degradation, implying that insufficient or excessive regularization may hinder model training.

TABLE IV
SIAMESE NETWORK PERFORMANCE UNDER DIFFERENT LABEL IMBALANCE MITIGATION STRATEGIES

Strategy	Accuracy	Precision	Recall
None	0.92	0.90	0.95
VAT	1.00	1.00	1.00
Oversampling ^a	0.98	0.97	0.98
Undersampling ^b	0.98	0.96	0.99

^a Oversampling balances label distribution by replicating minority class samples, mitigating bias toward majority classes.

^b Undersampling balances label distribution by reducing majority class samples, preventing dominance of majority classes.

TABLE V
ABLATION STUDY OF HYPERPARAMETERS (α_{VAT} AND α_{TL})

Accuracy	$\alpha_{\text{VAT}} = 0$	$\alpha_{\text{VAT}} = 0.2$	$\alpha_{\text{VAT}} = 0.4$	$\alpha_{\text{VAT}} = 0.8$
$\alpha_{\text{TL}} = 0$	0.92	0.94	0.97	0.94
$\alpha_{\text{TL}} = 0.2$	0.93	0.99	1.00	0.95
$\alpha_{\text{TL}} = 0.4$	0.98	1.00	1.00	0.96
$\alpha_{\text{TL}} = 0.8$	0.97	0.98	0.98	0.95

Fig. 9 presents the t-distributed stochastic neighbor embedding (t-SNE) visualization of features extracted by the CNN-based subnetwork under different loss function settings. These results, summarized in Table V, demonstrate that the complete loss function defined in this study effectively minimizes intra-class variation while maximizing inter-class separability. In contrast, removing either the triplet loss or the VAT loss results in less distinct clustering, underscoring the critical role of both components in improving feature discrimination and overall model robustness. These findings further validate the necessity of integrating multiple loss functions to achieve stable and generalizable feature representations.

B. Generalization Evaluation on Different Transformers

The first case study concerns a 500 kV transformer manufactured in September 2016 [1]. Routine FRA tests were conducted during commissioning, and the corresponding FRA fingerprints were recorded. Following a fault in April 2021, the transformer was disconnected from the grid, and subsequent FRA tests were carried out. Fig. 10a shows the FRA data

before and after the fault. Disassembly confirmed an IDSC fault in the LV c-phase winding. After preprocessing the FRA data, the trained Siamese network not only shows that it is dissimilar to normal data but also detects similarity with IDSC samples.

The second case study involves a 110 kV transformer manufactured in December 2010 [1]. FRA fingerprints were recorded during routine inspection, and in July 2017, the transformer experienced a fault requiring emergency repair. FRA tests were conducted during the repair process, and the corresponding data are presented in Fig. 10b. After preprocessing the FRA data, the calculated indicators are input into the trained Siamese network, which correctly identifies a DSV fault. The two case studies confirm that the trained Siamese network exhibits generalization, underscoring its potential for practical engineering applications.

C. Comparison with Previous Methods

This study compares the prior methods under data scarcity, as summarized in Table VI. Specifically, conventional FRA-based diagnostic approaches integrating SVM, transfer learning, and generative AI augmentation each exhibit certain strengths but remain constrained by limited generalization, reliance on simulated data, or instability of generated samples [1], [32], [37]. In contrast, the proposed method enhances sample diversity through random pairing and triplet loss in metric learning, while ensuring that all training pairs originate from real FRA data. Moreover, VAT effectively addresses the shortage of intra-class pairs introduced by random pairing. Furthermore, the combined use of triplet loss and VAT strengthens feature extraction and improves model robustness. Importantly, the framework is designed to evaluate similarity among FRA data, thereby adhering to the traditional comparison principle of FRA and ensuring interpretability, while also enabling the detection of previously unknown winding faults when the input data deviate from all baseline references.

V. CONCLUSION

This study proposes a metric learning framework integrating FRA to detect winding deformation faults under data scarcity. By reformulating the original dataset of N FRA samples into N^2 training pairs and further augmenting diversity through triplet loss (up to N^3), the method substantially expands the effective training set. VAT is incorporated to mitigate the inherent label imbalance between inter-class and intra-class pairs, while simultaneously promoting local smoothness

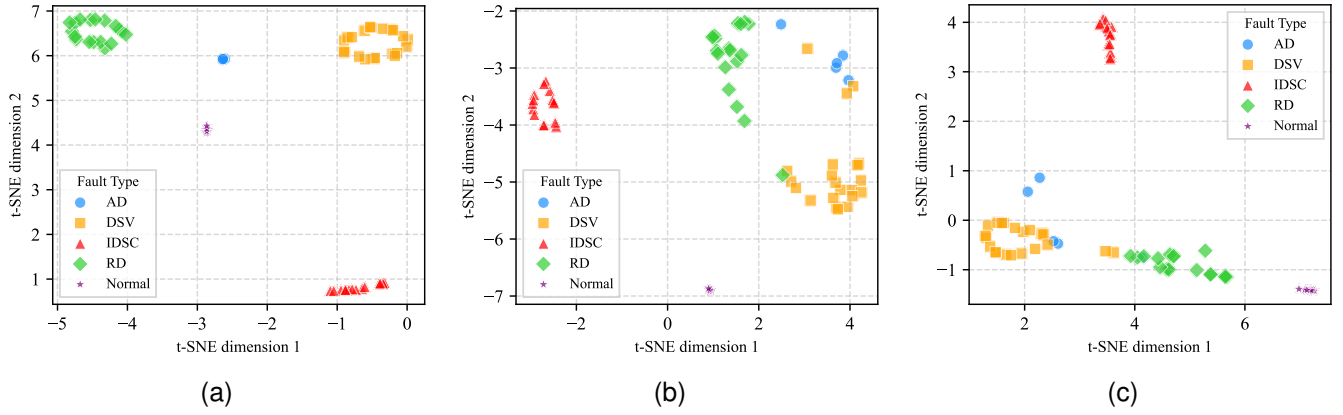


Fig. 9. T-SNE visualization under different loss functions. (a) Complete loss function. (b) Loss function without triplet loss. (c) Loss function without VAT loss.

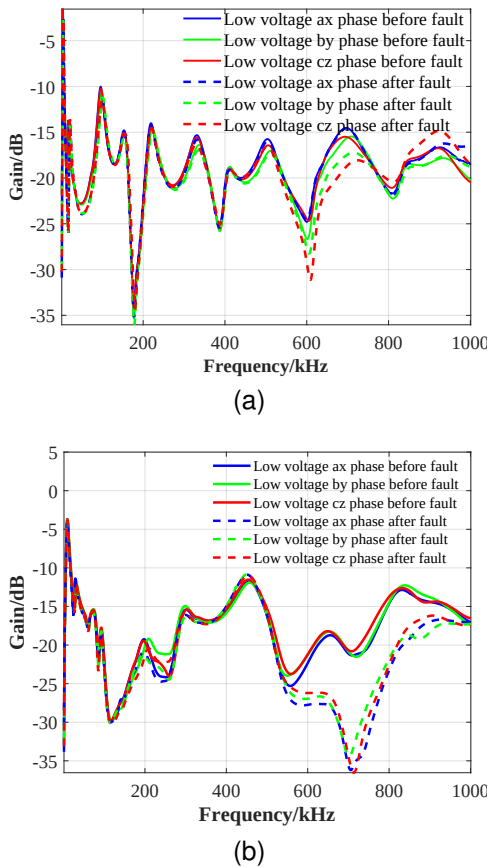


Fig. 10. FRA data before and after the fault [1]. (a) 500 kV transformer. (b) 110 kV transformer.

and enhancing robustness. Experimental results demonstrate that the proposed method achieves high diagnostic accuracy and strong generalization across different transformers, outperforming conventional SVM-, transfer learning-, and generative AI-based approaches. Consequently, the framework provides a practical, interpretable, and standardized solution for intelligent transformer fault diagnosis in scenarios with extremely limited labeled faulty data. Future work will extend the framework to multi-fault and cross-transformer cases, as

TABLE VI

COMPARISON OF DIFFERENT FRA-BASED TRANSFORMER FAULT DIAGNOSIS METHODS UNDER DATA SCARCITY

FRA + SVM [32]
• Effective for small-sample, nonlinear classification • Lower accuracy ($\approx 90\%$) • Limited generalization (single transformer) • Simple and direct • Requires manual feature engineering
FRA + Transfer Learning [37]
• Captures local correlations in FRA signatures • Transfer learning improves cross-transformer adaptability • High accuracy (up to 97%) • Heavily relies on simulation data • Accuracy drops in third transfer ($\approx 91\%$) • Computationally intensive
FRA + GAN-based Data Augmentation + SVM [1]
• Automatically expands limited FRA datasets • Improves robustness and generalization • Accuracy gain of $\approx 5\%$ over baseline • Generated data quality depends on GAN stability • Requires careful adversarial training tuning • Still limited by scarce real transformer data
Proposed Method
• All training pairs are derived from real FRA data • Perfect accuracy with VAT-balanced strategy • Strong generalization on real cases • Interpretable with clear engineering value • Potential to detect unknown winding faults • Slightly higher preprocessing complexity

well as integrate online FRA measurements for continuous condition monitoring.

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